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A New Methodology for Estimation of Israelis' International Migration by the Israel Central Bureau of Statistics (ICBS)[1]

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preface

International migration constitutes a principal part of the national statistics of most countries. Among the important tasks undertaken by the Israel Central Bureau of Statistics (ICBS) is definition and measurement of the population of migrating Israelis, in order to prepare population estimates, define the population of the Population Census, create sampling frames of populations for various surveys, and remove Israelis who emigrate from administrative files. Additionally, the migrant population is of great public and academic interest, resulting in a high demand for data and research on the subject.

In order to fulfil the required tasks and also maintain high statistical quality, the ICBS needs to produce reliable, homogeneous estimates of Israelis' international migration. In order to improve the national statistics on the subject in accordance with global changes in migration patterns and with technological advances, the ICBS set out to update the definitions of Israelis included in international migration statistics as well as upgrading to monthly estimates.

This methodological paper presents the background and rationale for developing the new methodology for measuring Israelis' international migration, as well as presenting the new definitions and data sources. Also presented are the limitations of the data, the new annual estimates, and a comparison with the former methodology (pre-change). It was found that the estimate of long-term Israeli emigrants is significantly higher using the new methodology (post-change), averaging 2.4 times higher than while using the former methodology. The estimate of returning Israelis was also found to be significantly higher using the new methodology, averaging 2.7 times higher than the estimate of returning Israelis using the former methodology. Additionally, the differences between population estimates and population growth rates calculated according to the two methodologies are presented.

Keywords: Emigration from Israel, International migration, Departing Israelis, Returning Israelis, Migration patterns, International migration balance, Population growth components, ICBS.

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Introduction

Reliable estimation of the extent of Israelis' international migration is crucial to the quality of the statistics produced by the Israel Central Bureau of Statistics (ICBS). International migration estimates play an important role in the CBS's fields of activity. Our work includes determining the de jure population living in Israel, for the following purposes: calculating population estimates; producing indices, samples, and statistical models; identifying and indicating the population not residing in Israel in administrative files from non-ICBS sources; and characterizing and researching various aspects of the population in Israel and the one emigrating from it.

Estimation of the migrants and their characteristics should reflect as much as possible the changing reality of migration patterns, and the extent to which trends among departing and returning Israelis can be identified and changes in their characteristics can be monitored in detail. In actuality, there is no single "correct" definition of who is a migrant from Israel, because (a) there is no survey that asks every departing or returning Israeli the purpose of their trip and (b) inaccuracies would certainly still occur as initial migration decisions can change with time. Therefore, the ICBS aims to adopt a demographic definition of emigration that includes as much of the population as possible who – with a reasonable level of certainty – moved their centre of life abroad and which, conversely, includes as little as possible of the population who did not. Due to these considerations, the ICBS decided to develop a new, updated methodology for estimation of international migration that would better meet the needs of national statistics. The results of this developmental process are presented in this article.

The aim of this methodological article is to compare the two methodologies (former and new) and to present the implications of the new methodology on population growth estimates. This is accomplished by presenting the methodology formerly used in the Population Sector of the ICBS for estimation of migrating Israelis, describing the rationale for updating it, followed by a presentation of the new methodology. Additionally, the data sources and the limitations of the new methodology are presented. The Results chapter presents the series of estimates obtained from using the new methodology and compares them to the estimates obtained using the former methodology. Additionally, the impact of implementing the new methodology on population estimates and on the rate of population growth will be examined.

Current Methodology for Estimating International Migration Among Israelis

The main methodology for calculating the flow of Israelis' migration movements has been used in its former form for several decades in the Demography Sector of the ICBS. The definitions guiding it have been in effect since 1995. The definitions of departing and returning Israelis were not symmetrical; that is, the number of days required to stay abroad for departing Israelis differs from the number of days required to stay in Israel for returning Israelis. **Departing Israelis** were defined as Israelis who stayed abroad 365 days or more continuously, and who before departing stayed at least ninety days continuously in Israel. **Returning Israelis** were defined as Israelis who stayed abroad 365 days or more continuously before entering Israel, and who after their return stayed at least ninety days continuously in Israel. In consequence, these definitions do not include Israelis residing abroad who visit their homeland for a short period (less than three months continuously) (CBS, 2023).

Since 2008, the source of data regarding departing and returning Israelis has been registration in the Border Control Registration system, which is conducted by border inspectors under the auspices of the Population and Immigration Authority. Border inspection stations are located at all entry/exit points on Israel's borders by air, land, and sea. In the past, when a resident/citizen exited or entered Israel, border inspectors documented the action by manually entering the person's information into the Border Control Registration system. Starting from 2010, passport information (without manual entry) has been scanned at Ben Gurion Airport, and by 2013, the scanning method was in use at all border stations in Israel except for a few border crossing points. Scanning is intended to reduce manual entry errors and thereby improve data quality. This system includes records of all exits/entries, the associated personal identification numbers, and the exit/entry dates. This data accumulates in statistical data files that are later completed with additional personal information from the Population Register: sex, date of birth, religion, date of immigration, marital status, and country of birth. The innovation since 2008 is that these records are transferred directly from the Border Control Registration system to the ICBS migration data system, and thus need less human intervention. The migration data system aggregates all movements of Israelis (including movements of East Jerusalem Arabs passing through international border crossing points, but excluding crossings between Israel and Palestinian Authority areas) at the borders in a calendar year (from January 1 to December 31 each year). The former methodology calculated the length of an Israeli’s stay abroad and their length of stay in Israel before departing. The system also performed the same calculations on the records of Israelis who

enter Israel. Calculation of departing flow (out-flow) and return flow (in-flow) is based on border movements only.

The products of the former system were two annual files: one for departing Israelis and one for returning Israelis (CBS, 2014b). In the ICBS's annual Statistical Abstract of Israel, annual data on Israeli residents who stayed abroad for 365 days or more continuously, and the returning Israelis from among these persons who stayed in Israel for at least ninety days continuously, are published according to sex and age. The ICBS also publishes a media release each year t entitled "Departures and Returns in [year] (e.g., 2020) of Israelis Staying Abroad Continuously for a Year or More", which contains estimates for year t-2 of the number of departing and returning Israelis, their demographic characteristics, and an estimate of the stock of Israelis residing outside Israel.

Motivations for Updating the former Methodology

In recent years, there has been growing recognition within the ICBS's Senior Department of Demography and Census of the need to update the statistics of international migration through the development of an up-to-date methodology that is more suitable for its aims. In particular, the need arose to update the definitions of departing and returning Israelis such that these reflect the changing reality of migration patterns in the present times. These patterns are influenced by the increased availability and decreasing costs of international travel, the rapidly developing connections of Israel to the global economic market, increased business opportunities abroad, the development of the IT sector leading to job opportunities requiring relocation and study abroad, and the expansion of the dual citizenship phenomenon, along with an Israeli society characterized by close familial and social ties.

These factors have led, on one hand, to an increase in the number of departing Israelis and, on the other hand, to an increased tendency to return for visits to Israel. Therefore, the former methodology, which requires a stay abroad of a full continuous year (if this sequence is 'broken' by a visit to Israel, the person will not be considered an Israeli emigrant), is less suited to contemporary migration patterns. Additionally, there is no usage of the stock of Israelis residing abroad in determining whether a person crossing the border belongs to the out-flow or the in-flow migration movements. This can lead to a person being registered as a departing Israeli twice without being counted as a returning Israeli between one departure and the next). Moreover, there is no integration of information regarding the person's civil status in the Population Register, as a result of which persons registered in the Population Register who have never resided in Israel might be included in the estimates as migrants.

A team was formed within the ICBS's Senior Department of Demography and Census which aim was to conducted a research and based upon to formulate recommendations for improving and updating the handling of international migration statistics. The recommendations of the team were (1) to establish one comprehensive "emigrant" stock to serve all of the ICBS's needs, and (2) that this stock should include two main components: (a) a list of Israelis (stock) at a specific point in time (an exact date) who are highly likely not to belong to the population of Israel at that time; and (b) a multi-year stock of flows including a list of people who emigrated and/or returned, with information necessary for managing their movements. The definitions of departing and returning Israelis would take into account short visits in Israel, and would provide information about short-term and long-term emigrants. The operational distinction between the needs of the Population Census and the needs of the Demography Sector were preserved in the recommendations.

At the beginning of the development of the new methodology, five alternatives for defining departing Israelis and returning Israelis were researched, differing in various combinations of three guiding principles: (a) the existence or non-existence of a required initial duration of continuous stay abroad; (b) the cumulative number of days spent abroad; and (c) maintaining symmetry or allowing for asymmetry in the definitions of departing and returning Israelis. The alternatives were examined based on data from border crossings and calculations of Israelis' stays in Israel and abroad in a closed population (without taking into account natural increase or the arrival of "olim" (immigrants to Israel under the Law of Return) for 2009–2013. The characteristics of the Israelis included in each alternative were examined through use of their demographic and socio-economic characteristics.

The findings of the research recommended the methodological alternative presented later in this paper, which implements the goal of including as much as possible the population that has likely relocated its centre of life outside of Israel and excluding as much as possible the population that has not done so, compared to other alternatives. Furthermore, this alternative supports the need for an initial period of a continuous stay of ninety days abroad, while allowing for an additional period of cumulative days abroad, totalling at least nine months out of the twelve months since departing. In addition, this alternative maintains symmetry between the definitions of those departing and those returning. The research findings and UN recommendations were adopted by the advisory committee on demographic issues of the Public Advisory Council for Statistics, which included demography experts headed by Dr. Eliyahu Ben-Moshe (chairman of the advisory committee) and the former National Statistician Professor Danny Pfeffermann, and the development of the chosen methodology was initiated.

The New Methodology for Estimating International Migration Among Israelis

The population included in the new methodology was determined according to the definitions of the de jure population, upon which population estimates have been based by the ICBS since the 2008 census. The de jure population includes the total number of Israelis registered in the Population Register having the status of citizen, permanent resident, or temporary resident. However, there are persons registered in the Population Register who have never resided in Israel for an extended period, such as immigrating citizens (a person born to an Israeli citizen while residing abroad). These persons should not be included in estimates of departing Israelis because they were never part of the de jure population in Israel in the first place. These immigrating citizens, as well as new registrants in the Population Register with an initial status of not an active resident are not included in the estimate of the flow of departing Israelis, but are included in the gross stock of Israelis staying abroad (detailed explanation of the stock is provided later in the paper).

The new definitions established are as follows:
Israeli emigrants: An Israeli who stayed abroad for at least 9 months cumulatively during the year following their departure, of which the first 3 months were continuous.
Returning Israeli immigrants: An Israeli who stayed in Israel for at least 9 months cumulatively during the year following their entry, of which the first 3 months were continuous. This indicates an Israeli who returns to live in Israel after emigrating.

These definitions are based on establishing a transfer of one's centre of life and on a substantive cumulative stay abroad versus measurement of just continuous residence in a foreign country. They are measured symmetrically for departing and returning Israelis, thus providing a uniform, equal-opportunity criterion for inclusion in the migration flows and preventing multiple or conflicting classifications. Additionally, the initial required threshold of ninety continuous days abroad or in Israel serves to distinguish between a tourist and a migrant (a regular tourist visa is usually granted for a period of up to ninety days).

In addition to the updates of definitions for Israeli emigrants and Returning Israeli immigrants, we created two additional types of migration flows that did not exist previously: **Short-term emigrants**, referring to persons who stayed abroad for between 3 months (initially and continuously) and up to 9 months (cumulatively) during the year following departure. **Visiting Israelis** refers to Israelis residing permanently abroad who came for a relatively prolonged visit in Israel, staying in Israel for between 3 months (initially and

continuously) and 9 months(cumulatively) during the year following entry. In addition to the migration flows mentioned above, the first population stock of Israelis residing abroad was created.

The stock of Israelis residing abroad is based on the results of the Population and Housing Census 2008, which determined who belonged to the population residing in Israel and who did not, at the end of 2008. From 2009 onward, the stock is updated according to the migration flows into and out of the country, and according to population status updates as recorded in the Population Register. Contributions to the stock include: (a) Israeli emigrants (see above definition) and (b) Israelis registering for the first time in the Population Register whose status is " not active", or who are "active" but have never resided in Israel for a long period of time. Removal of records from the stock include: (a) Returning Israeli immigrants (see above definition), (b) Death notifications of persons included in the stock as they are recorded by the Population Register, or (c) persons included in the stock reaching the age of 110, whose death was not officially reported. This gross stock allows administrative indication for each person in the Population Register of whether, or not, they are part of the population residing permanently in Israel at a certain point in time. However, this stock is still an overestimation of the number of Israelis emigrants living abroad, because it includes an unknown number of deceased persons who have died aboard without their death being reported to the Population Register; as well as children of Israeli citizens born abroad who are registered in the Population Register without ever been permanent residents of Israel.

Sources of Information for the New Methodology

The quality of international migration statistics derives from the quality of the data sources available to formulate these statistics. The more comprehensive, reliable, and up-to-date a data source is, the higher the quality of the data and thus the ability of statisticians, decision-makers, researchers, and the general public to rely on them.

The main data source for formulating the new methodology is the border control system under the auspices of the Population and Immigration Authority. Israel has a unique situation in that its borders are relatively controlled and closed compared to other countries. In other words, people cannot move freely between Israel and neighbouring countries without registering their movement at land border crossing points (the situation differs in most European countries, for example), and registration of movements at non-land border crossings is done through airport and seaport crossing points. This situation allows almost complete coverage and documentation of all movements by residents across the borders. The data is gathered via automatic

computerized scanning of passports conducted at all border stations and crossing checkpoints and transmitted directly to the ICBS.

The Population Register serves as a **secondary data source**. A person's citizenship status upon their first registration in the Population Register is obtained from this source. An initial registration of an Israeli in the Population Register with a status of "not active" (according to the Population Authority's discretion), or that of an immigrating citizen who resides abroad, indicates that the person is not actually residing in Israel, but that there is a future possibility that they will come to reside in Israel if they so wish. These persons are added to the stock of Israelis residing abroad. In addition, the Population Register serves as a source for receipt of notifications of the deaths of Israeli residents living abroad, thus allowing their removal from the stock of Israelis staying abroad. This source provides complete coverage of the new registration of persons in the Population Register and partial coverage of death notification for those residing abroad.

Data Limitations

There are several systemic limitations to the obtained data which might affect the data quality, but their impact on the overall data quality is marginal. The limitations are as follows:

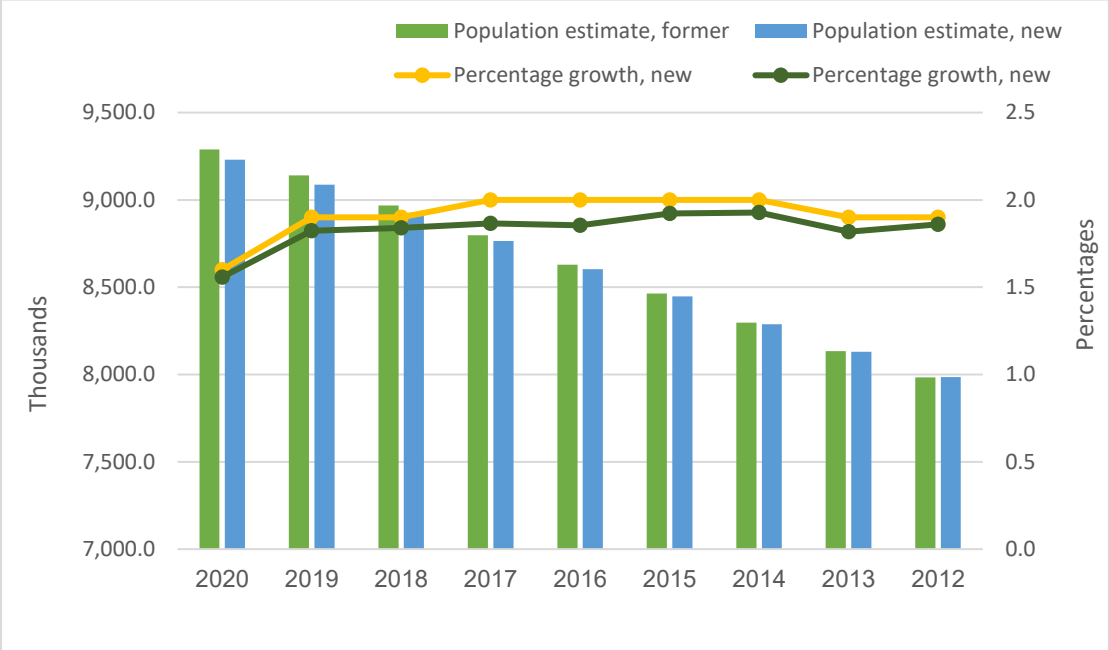
- Use of a foreign passport when exiting or entering, for someone with dual citizenship, might lead to incomplete registration of a person's border movements. Since 2008, the new border control system should prohibit Israeli passport holders from exiting or entering Israel using a foreign passport. Thus, if a border crossing with a foreign passport occurs, the system is programmed to link it to their Israeli identity information. However, it could occur in some rare cases this linkage is not successful.
- If the death of an Israeli who was included in the stock of Israelis residing abroad is not reported by the family to the Population Register in Israel (probably more common in cases, in which there is no incentive or benefit to report it), this person will not receive a status of deceased in the Population Register, and therefore will not be removed from the stock. Thus, the gross stock of residents abroad is an overestimate for the number of Israelis residing abroad. A certain amount of correction is done by removing anyone who has reached the age of 110 (by given the status of "presumed deceased").
- It might be somewhat problematic that our determination of whether a person is a departing Israeli, or a returning Israeli, is dependent on their presence or lack thereof, in the stock of residents abroad. This applies in cases in which the initial determination of place of residency in the

- 2008 census was erroneous. If so, the error will be carried over to future years if the person does not cross any of the borders thereafter.
- Because the new migration definitions are general and are applied equally and objectively to the entire population, without considering individual travel patterns, it is possible to incorrectly classify persons as non-emigrants even if they have relocated their centre of life abroad (false negative). This situation can occur if the number of days spent by a person abroad does not meet the definition set for Israeli emigrants, and therefore they will not be classified as such. For example, persons who consider themselves residents of another country, but who have never stayed 3 months continuously abroad or who divide their time fairly equal between abroad and Israel, will not be defined as emigrants under the new definitions. The same might happen with returning Israeli immigrants.
- The data include international border crossings of Israel for Israelis (including East Jerusalem residents), and exclude crossings of Israelis into the Judea and Samaria Area and the Gaza Strip, and vice versa.

Examining the impact of transitioning to the new methodology on population growth and its estimates

In order to assess the extent of the impact of transitioning to the new methodology on the components and rate of population growth, and thus on the population estimates of Israel, the population estimate was calculated in two different ways: (a) based on the components of growth under the former methodology as published in the ICBS's annual Statistical Abstract; and (b) based on the same components of growth (births, deaths, "olim" - immigrants, immigrating citizens, and family reunification), excluding the international migration balance component of Israelis, which was recalculated according to the new methodology. The simulation was conducted for 2012–2020. Additionally, it should be noted that the components of growth for 2020 were exceptional because they were influenced by the global COVID–19 pandemic, which affected persons' ability and/or willingness to immigrate to and emigrate from Israel, as well as resulting in a decrease in the number of births and an increase in the number of deaths in that year. The following diagram illustrates the results of the population calculations using both methods.

Diagram 1. Population estimate and percentage population growth according to the former and new methodologies, 2012–2020



An analysis of the results shows that, as expected, the use of the new methodology reduces the population estimate compared to the former estimates. Of the population components, the natural increase balance component accounts for approximately 83% on average annually for the years 2012–2020 and the international migration balance component accounts for approximately 17% of the total annual population growth in the former methodology. In contrast, in the new methodology, the natural increase balance component increases to approximately 86% on average, and the international migration balance component decreases to approximately 14% on average. The contribution of the international migration balance component to population growth decreases by 4% on average in the transition from the former to the new methodology, and the contribution of the natural increase balance to the population growth increases by an equal percentage.

Because the data on natural increase, immigrants, returning citizens, and family reunification were held constant, the reduction in the population estimate stems from the new definitions of departing and returning Israelis in the new methodology. This mainly results from a larger increase in the number of departing Israelis compared to a smaller increase in the number of returning Israelis. The new, more flexible and symmetric definitions increase the negative

migration balance between out-flow and in-flow, which in turn reduces the overall international migration balance of Israel, though it remains positive. The reduction in the international migration balance is also reflected in the slowdown of population growth under the new methodology; the population decreased by approximately 0.1% per year compared to the growth rate under the former methodology.

The decrease in the growth rate of the total population is not surprising because of the broadening of the definitions of who is considered an Israeli emigrant. This happens by allowing for cumulative stays abroad, which translate into allowing visits to Israel during the first year after departing, rather than requiring a stay of a consecutive year abroad. This in turn leads to a higher estimate of departing Israelis, thus reducing the share of the international migration balance component in the overall population growth of Israel. This decrease in the total population growth rate is not dramatic because Israel's population growth rate is mainly influenced by the natural increase. Nevertheless, it should not be overlooked that there is a cumulative effect: The decrease in the growth rate added up to approximately 4.2% cumulatively during 2012–2020. Adding up all the differences between the negative migration balance (more emigrating than returning Israelis) under the new methodology compared to this negative balance under the former methodology results in a cumulative population loss of approximately 57,500 persons for 2010–2020.

Summary

The paper highlights the need to update definitions of departing Israelis, by allowing visits to Israel within the first year after departing abroad, and by maintaining symmetry by using the same definitions among returning Israelis. Numerous studies conducted by the ICBS indicate that updating the definitions of departing and returning Israelis will help produce more reliable statistics regarding the identification of Israelis and who moved their centre of life abroad and those who decided to return to Israel. This will contribute to the calculation of more reliable estimates of the population residing permanently in Israel which will enable better-informed policy planning based on this.

Additional benefits of the new methodology include creating new immigration statuses of short-term emigrants (from three to nine months cumulative stay abroad) and of visiting Israelis (from three to nine months cumulative stay in Israel), along with the existing statuses of long-term emigrants and of returning Israelis. The symmetry between these definitions facilitates comparability and eliminates the possibility of overlapping immigration statuses in the same year. Furthermore, establishing a database of Israelis residing abroad based on migration flows enables the identification

of persons registered in the Population Register who are currently residing abroad at a specific point in time, thus enabling the classification of Israelis in the Population Register into three primary groups: persons residing in Israel, persons residing abroad, and deceased persons. The latter two groups should be removed from calculations of population estimates, statistical indices, and when sampling populations for surveys, as well as being removed from administrative files received from data sources outside the ICBS.